

A Resource-Efficient Federated Learning Framework for Intrusion Detection in IoMT Networks

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Abstract—The emergence of medical sensors in Smart Healthcare Systems (SHS) has enabled the development of sophisticated Internet of Medical Things (IoMT) networks which is crucial for tracking vital physiological parameters in consumer electronics. These networks are progressively incorporated into consumer devices allowing users to monitor their health parameters effortlessly. Nonetheless, they encounter considerable security and privacy issues stemming from weaknesses in data transfer. Despite the potential of Intrusion Detection Systems (IDS), there is a critical need for a real-time, highly precise attack detection system optimized for the edge-centric Internet of Medical Things (IoMT) environment in consumer sensor devices. Therefore, we introduce the federated kolmogorov-arnold network (FedIoMT), an advanced federated learning framework that utilizes meta-learning along with an advanced clustering method to achieve robust model aggregation. Our FedIoMT acts as both an adaptive personalized updating mechanism and a shared global classifier. In addition, we propose a novel intrusion detection model that uses the kolmogorov-arnold convolutional network (KANConvNet) as its local classifier, which improves scalability and interpretability in the FL framework updates. Our FedIoMT model demonstrates superior performance compared to other existing FL-based methods, achieving validation accuracies of 99.38%, 99.23%, 99.26%, and 99.2% across four benchmark IoMT datasets under the IID setting. FedIoMT utilizes significantly more floating point operations per second (FLOP) counts up to 2.8 times more than other models and also excels in memory efficiency, consuming up to 93% less peak memory across devices compared to the alternatives. We thoroughly evaluated FedIoMT on embedded consumer devices, demonstrating its resource-efficient characteristics.

Index Terms—Internet of Medical Things, intrusion detection, federated learning, Kolmogorov–Arnold networks. FedIoMT.

I. INTRODUCTION

THE Internet of Medical Things (IoMT) connects smart medical devices such as blood pressure monitors and glucose meters to consumer electronics found in wearables and

home healthcare goods. Because of the sensitivity of the data being communicated, these devices, which account for a 100 billion market, require secure connection. [1], [2], [3]. With the increasing use of Wi-Fi, 5G, and NB-IoT in consumer electronics for data transmission, stronger security measures are essential to ensure confidentiality, integrity, and availability [4], [5]. Data storage on cloud gateways within these consumer-driven healthcare ecosystems further emphasizes the need for robust protection mechanisms [6]. The rapid expansion of IoMT in consumer electronics, combined with resource constraints and data diversity, makes it a prime target for attacks, compromising the security of sensitive medical information and threatening patient safety [7], [8], [9], [10], [11], [12].

Recent research indicate that 53% of Internet of Medical Things (IoMT) devices integrated into consumer electronics exhibit substantial susceptibility to hackers, hence presenting considerable threats to interconnected healthcare systems [9], [13]. An examination of 40 IoMT-enabled consumer products identified 18 unique attack vectors, highlighting the diverse risks aimed at healthcare technology in consumer electronics [14]. Ransomware assaults have increased by more than 50%, resulting in significant operational disruptions in consumer-integrated healthcare devices and jeopardizing patient safety [15]. Prevalent cyber threats, such as DoS, DDoS, MitM, and Ransomware, threaten the confidentiality, integrity, and availability of critical medical consumer products like pacemakers, insulin pumps, and smart wearables. These vulnerabilities have a substantial influence on healthcare delivery in consumer electronics environments [16], [17], [18], [19]. Figure 1 depicts prevalent threats to IoMT devices, including Denial of Service (DoS), Distributed Denial of Service (DDoS), Man in the Middle (MitM), and Ransomware.

To improve network security in consumer electronics, particularly IoMT devices, intrusion detection systems (IDS), alongside machine learning (ML) and deep learning (DL) techniques, are widely employed. IDS leverages heuristic methods and fuzzy logic to thwart attackers and safeguard IoMT networks [19]. Sajid et al. [20] proposed a hybrid model combining XGBoost and CNN for feature extraction, with LSTM optimized for security. Turukmane and Devendiran [21] introduced M-MultiSVM with Mud Ring optimization for hyperparameter tuning in IoMT-based

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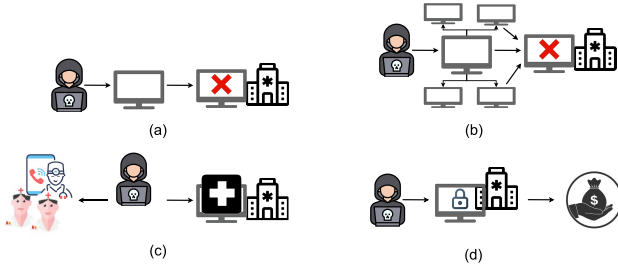


Fig. 1. Examples of security threats (a) DoS (b) DDoS (c) MitM (d) Ransomware to IoMT.

security systems. Talukder et al. [22] integrated ML with SMOTE-TomekLink for more effective intrusion detection. Goswami et al. [23] developed a Cu-BLSTM-based IDS to detect various attacks in IoMT. Chen et al. [24] proposed a hybrid feature selection model combining chi-square and information gain with ML/DL algorithms for enhanced protection. Despite these advancements, IoMT devices still encounter challenges like limited processing power, diverse environments, and inadequate labeled data [24].

From a data privacy perspective, sharing network data containing sensitive patient information is challenging for data owners, such as hospitals [25]. This makes developing effective IDS schemes difficult. Federated learning (FL), proposed by Google in 2016, addresses this by enabling ML on distributed data without sharing sensitive information [26]. FL enhances IoMT security by improving threat detection and resilience against cyber-attacks through decentralized model training, preserving privacy while maintaining high accuracy [27].

In federated learning, FedAvg and FedProx have different implications for model performance at the local and global levels [28], [29]. FedAvg works well in IID settings but struggles with non-IID data, as it averages local model updates without accounting for data heterogeneity, leading to slower convergence and potential bias in the global model. In contrast, FedProx introduces a regularization term that stabilizes local updates, improving convergence and robustness in non-IID environments by preventing local models from deviating too far from the global model. This makes FedProx more effective in heterogeneous data scenarios, resulting in more accurate and stable global model performances [30].

However, these FL architectures are vulnerable to security threats, especially with heterogeneous data, as adversarial clients can degrade global model performance through attacks like A Little Is Enough (ALIE) attack and Inner Product Manipulation (IPM) [27], [31], [32]. While robust aggregation methods exist, they suffer against sophisticated attacks, particularly in Non-IID consumer devices. Defense measures often discover threats too late and some, like FLTrust, undermine privacy by demanding server access to sensitive data [31], [34], [35].

The study presents FedIoMT, a Federated Learning (FL) methodology tailored for IoMT devices in consumer electronics, tackling issues including data heterogeneity and model modification via multi-center clusters and similarity analysis. FedIoMT utilizes meta-learning to enhance the global

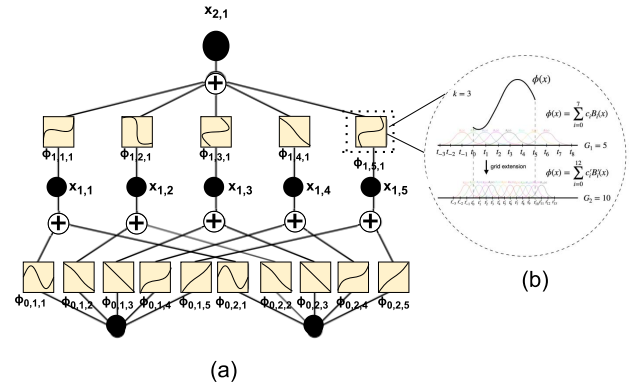


Fig. 2. On the left (a) the activations as they progress through the network. On the right (b) the activation function is represented using a B-spline, enabling the adjustment between coarse-grained and fine-grained grids.

model and facilitates individualized sharing with individual devices. We introduce KANConvNet, derived from the Kolmogorov–Arnold Network (KAN), which customizes activation functions to accommodate the distinct data of each IoMT device, as depicted in Figure 2. Integrating KAN layers into a CNN framework, KANConvNet effectively captures intricate data patterns, enhancing efficiency and robustness within the FedIoMT architecture. Therefore our system model is crucial for consumer electronics and healthcare. Therefore the main contributions are listed as follows:

- To the best of our knowledge, we introduce the KANConvNet model first time within the FedIoMT framework, which uses the Kolmogorov–Arnold theorem and tailored activation functions in a convolutional structure to improve scalability, robustness and feature interaction in intrusion detection.
- Our FedIoMT model incorporates a meta-learning phase to enhance security and counter adversarial clients using the TOPSIS method. This approach leverages clustered local models and ensures KRUM’s robust aggregation of similar models tailored to various consumer electronics devices. Additionally, we evaluate the performance using the Multi-Krum aggregation strategy for comparative analysis, achieving improved computational efficiency and enhanced robustness.
- We perform a comprehensive design space search with a genetic algorithm (GA) to optimize hyperparameters and improve model performance though fine tuning the process.
- FedIoMT achieves high efficiency, reducing peak memory usage by 73.3% to 93.8% while maintaining a high accuracy of 99.37%, making it suitable for consumer devices with limited memory.

Following is a synopsis of the remainder of this paper. In Section II, we introduce our connected work with the previous studies comparing our proposed technique in this domain. In Section III, we address the approach we discuss our data collection, preprocessing methods. In Section IV, we describe our proposed method defining system initialization. In Section V, we mainly focus our optimization process using selected hyperparameter design exploration through genetic

algorithm. In Section VI, we define our experiment setup and result analysis defining state of the art with our proposed model. In Section VII, we finally conclude our work defining contributions and limitations of our work.

II. RELATED WORK

Given the limited research on intrusion detection in IoMT within consumer electronics, this section mainly provides a concise overview of relevant studies focused on intrusion detection, emphasizing Federated Learning (FL)-based methods and Personalized Federated Learning (PFL).

A. Intrusion Detection Schemes

Intrusion detection is crucial for network security, especially in smart healthcare and IoMT. Shambharkar and Sharma [36] developed a deep learning model combining CNN and LSTM with attention mechanisms to detect intrusions by processing 100-dimensional feature vectors. Bouke et al. [37] improved IoMT intrusion detection using the WUSTL EHMS 2020 dataset, employing SMOTE-ENN for data balancing and a sequential deep learning model for high accuracy. Faruqi et al. [38] proposed a hybrid CNN-LSTM model using CIC-IDS2017 and Maling datasets for feature extraction and data normalization. Shambharkar and Sharma [36] introduced LinSVM, ConvSVM, and CatEmb models for network-based attack detection, utilizing feature scaling and SMOTE. Anjelin and Kumar [39] optimized intrusion detection with an Enhanced Elephant Herding Optimization (EHO) algorithm and CNN, achieving high accuracy for IoMT environments.

B. Federated Learning-Based Intrusion Detection Methods

FL has been widely utilized across several domains to address the challenges of data isolation and privacy protection in consumer electronics. When applied to intrusion detection in practical circumstances, FL has demonstrated promising results. Javeed et al. [40] proposed a hybrid FL model combining CNN and BiLSTM for intrusion detection, leveraging spatial and temporal features while maintaining data privacy through a zero-trust approach. Olanrewaju-George and Pranggono [41] developed an FL-based IDS for IoT, utilizing both unsupervised and supervised deep learning models, with the FL AutoEncoder outperforming non-FL models. Alazab et al. [42] introduced a privacy-preserving IDS using FL with the FedAvg method, enabling collaborative model training without data sharing. Rashid et al. [43] used a hybrid FL model with Stacked CNN and BiLSTM to detect complex threats in WSNs, enhancing privacy. Sarhan et al. [44] proposed a FL-based CTI sharing scheme for NIDS, allowing collaborative model training across organizations without exposing sensitive data.

However all existing studies mostly employ the FedAvg algorithm to improve intrusion detection models, but frequently neglect the difficulties posed by non-IID intrusion data in real-world situations. Moreover, these FL frameworks are susceptible to model poisoning assaults that undermine performance. Existing models have challenges when using a method that calculates global averages. In order to tackle these

problems, our FL system incorporates a multi-center server that facilitates customizable model sharing. Additionally, we utilize the KANConvNet model to enhance generalization and reliability on consumer driven devices.

The principal obstacles in the distributed systems in ensuring accurate weight updates in decentralized models are data heterogeneity. Besides, it often face problems with restricted computational resource. The FedIoMT architecture tackles these difficulties through multiple techniques. It utilizes TOPSIS to mitigate the influence of low-scoring clients during meta-learning, ensuring that contributions from less dependable clients do not compromise the integrity of the global model. Besides, our Krum-based aggregation guarantees reliable model updates by identifying the most trustworthy client models and excluding those with poor performance or anomalous data. These solutions collectively enhance the efficiency, robustness, and accuracy of decentralized weight updates, hence improving the overall performance of FedIoMT.

III. DATA COLLECTION AND PREPROCESSING

This section discusses the dataset, preprocessing steps, data splitting, normalization, and feature selection. The WUSTL-EHMS-2020 dataset [45] was used, containing 16,318 samples with 87.5% normal and 12.5% attack data. Additional datasets, including ECU-IoHT [46], BlueTrack [47], and RT-IoT [48], were employed for benchmarking. The data was split into 60% for training, 20% for local testing, and 20% for global testing, with the global test data containing unknown attack types. To enhance model convergence and reduce training time, min-max normalization was applied. Due to class imbalance, SMOTE-ENN [50] was utilized for data balancing, generating synthetic samples to improve model validation and training without losing data through undersampling. Feature selection was based on information gain [49], with features reducing uncertainty in the target variable being retained, and a threshold value of 0.05 was used.

IV. PROPOSED METHOD

In this section, we explain our FL system, designed specifically for IoMT to enable safe and fast learning across distributed consumer devices. Our FedIoMT model trains locally using KANConvNet within convolutional layers for better local optimization. Furthermore, in the case of dimensionality reduction and ranking, we utilize the TOPSIS method for ranking and clustering for similarity analysis along with outlier reduction of the updated gradients for the aggregation. Afterward, we implement a robust aggregation approach based on using Krum to protect against adversarial threats. The proposed workflow is depicted in Figure 4. Finally, we show the KANConvNet intrusion detection model as local model for different consumer sensor based devices for IoMT.

A. System Initialization

The FL process starts with setting up the server and IoMT devices as clients, involving model initialization, communication setup, and security protocols to ensure data privacy and integrity.

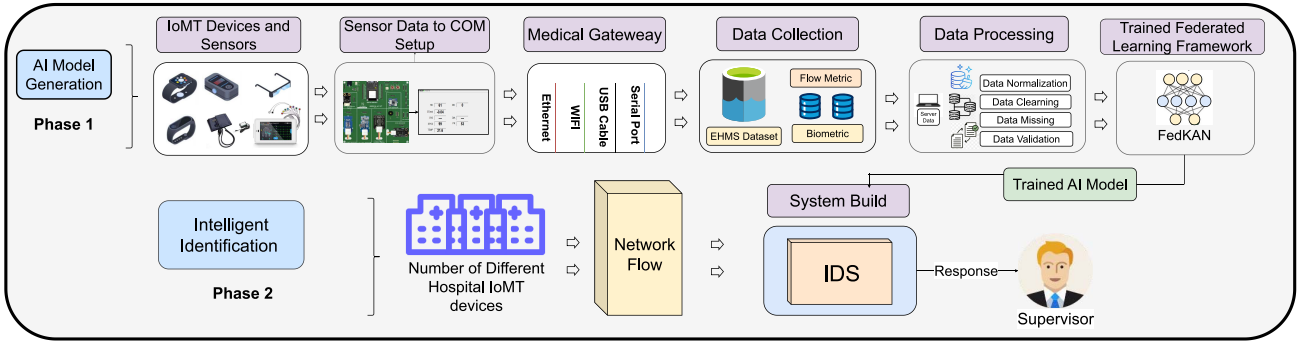


Fig. 3. Construction Phases of the IDS based on our Proposed FedIoMT Model Architecture.

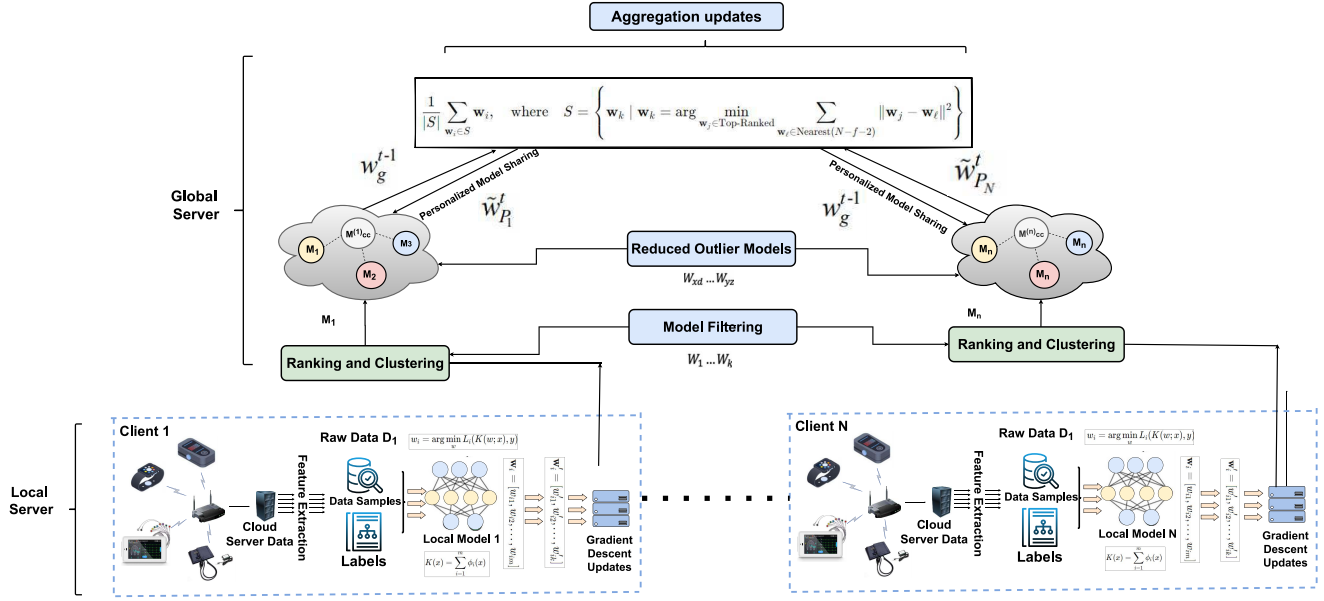


Fig. 4. Our Proposed FedIoMT Model Workflow.

1) *Model Initialization*: The central server initializes the global model $\mathcal{M}_G^{(0)}$ with pre-trained weights, which is distributed to all IoMT devices D_i for $i = 1, 2, \dots, N$:

$$\mathcal{M}_G^{(0)} \rightarrow D_i, \quad \forall i \in \{1, 2, \dots, N\}, \quad (1)$$

2) *Communication Setup*: A secure communication protocol $\text{Secure_Channel}(\text{Server}, D_i)$ is established for each IoMT device D_i , ensuring protected exchange of model updates and preventing unauthorized access:

$$\text{Secure_Channel}(\text{Server}, D_i), \quad \forall i \in \{1, 2, \dots, N\}, \quad (2)$$

B. Global Model Update

Local model updates are aggregated into multi-center clusters using a resilient mechanism to form the new global model $\mathcal{M}_G^{(t)}$:

$$\mathcal{M}_G^{(t)} = \text{Aggregate}(\{\mathcal{M}_i^{(t)}\}_{i=1}^N), \quad (3)$$

This process accounts for potential adversarial updates and is robust against attacks. The updated global model is then distributed to all IoMT devices, ensuring synchronous communication and coordinated learning across the network.

C. Local Optimization

Each IoMT device D_i optimizes the global model locally by minimizing its objective function $\mathcal{L}_i(\mathcal{M}_i^{(t)})$, defined as:

$$\mathcal{L}_i(\mathcal{M}_i^{(t)}) = \frac{1}{|\mathcal{D}_i|} \sum_{(x,y) \in \mathcal{D}_i} \ell(f(x; \mathcal{M}_i^{(t)}), y), \quad (4)$$

where $\ell(\cdot, \cdot)$ is the loss function, $f(x; \mathcal{M}_i^{(t)})$ is the model's prediction for input x , and $|\mathcal{D}_i|$ is the size of the local dataset.

1) *Gradient Descent-Based Optimization*: The local optimization phase fine-tunes the global model to each IoMT device's specific data using gradient descent:

$$\mathcal{M}_i^{(t)} \leftarrow \mathcal{M}_i^{(t-1)} - \eta \nabla \mathcal{L}_i(\mathcal{M}_i^{(t-1)}), \quad (5)$$

where η is the learning rate and $\nabla \mathcal{L}_i(\mathcal{M}_i^{(t-1)})$ is the gradient of the local loss.

2) *Personalization and Regularization*: To address data variability among IoMT devices, each device applies L2 regularization to avoid overfitting and leverage prior knowledge. The regularized objective function is:

$$\mathcal{L}_i^{\text{reg}}(\mathcal{M}_i^{(t)}) = \mathcal{L}_i(\mathcal{M}_i^{(t)}) + \lambda \Omega(\mathcal{M}_i^{(t)}), \quad (6)$$

where λ is the regularization parameter and $\Omega(\cdot)$ represents the regularization term.

D. Local to Global Model Transmission

After local optimization, each IoMT device sends its updated model $\mathcal{M}_i^{(t)}$ to the multi-center servers.

The personalized update algorithm in our FL framework consists of two main phases: Dimensionality Reduction using TOPSIS and Robust Aggregation using Krum. These phases ensure tailored updates for each device while maintaining resilience against adversarial attacks.

1) *Dimensionality Reduction Using TOPSIS*: We utilize the TOPSIS approach [55] that is mainly employed by the central server to minimize the dimensionality of the model updates received from the IoMT devices. This stage is critical for addressing the high dimensionality and variety of data across different devices. The procedure involves the following steps:

a) *Normalization*: The server first normalizes the model updates $\mathcal{M}_i^{(t)}$ from each IoMT device D_i . The normalization ensures that the models are comparable across different scales. Mathematically, the normalized value $m_{ij}^{(t)}$ for each model parameter j of device D_i at round t is calculated as:

$$m_{ij}^{(t)} = \frac{m_{ij}^{(t)}}{\sqrt{\sum_{i=1}^N (m_{ij}^{(t)})^2}}, \quad (7)$$

where $m_{ij}^{(t)}$ is the j -th parameter of the model update from device D_i at round t , and N is the total number of devices.

b) *Weighted normalization*: After normalization, the server applies a weighted normalization to account for the relative importance of different model accuracy values as parameters. The weighted normalized value $w_{ij}^{(t)}$ is computed as:

$$w_{ij}^{(t)} = w_j \cdot m_{ij}^{(t)}, \quad (8)$$

where w_j is the weight assigned to the j -th parameter. These weights are determined based on the relevance of each parameter to the overall model performance.

c) *Ideal and anti-ideal solution identification*: The server identifies the ideal solution A^+ and the anti-ideal solution A^- . The ideal solution represents the most desirable set of model parameters, while the anti-ideal solution represents the least desirable set. These are calculated as:

$$A^+ = \left\{ \max_i w_{ij}^{(t)} \mid j = 1, 2, \dots, d \right\}, \quad (9)$$

$$A^- = \left\{ \min_i w_{ij}^{(t)} \mid j = 1, 2, \dots, d \right\}, \quad (10)$$

where d is the total number of parameters in the model.

d) *Similarity calculation*: The server computes the similarity of each model update to both the ideal and anti-ideal solutions. The distance of each model updates D_i to the ideal solution A^+ and the anti-ideal solution A^- is given by:

$$S_i^+ = \sqrt{\sum_{j=1}^d (w_{ij}^{(t)} - A_j^+)^2}, \quad (11)$$

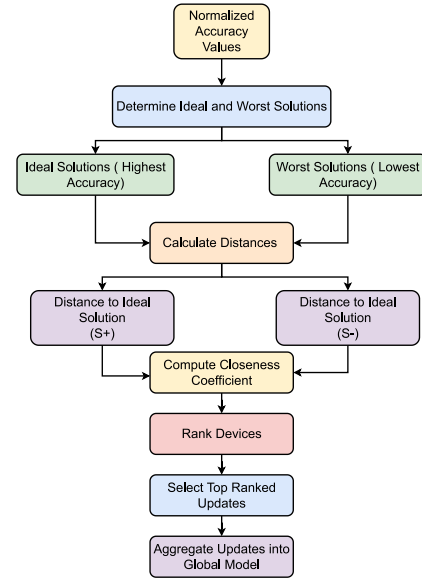


Fig. 5. TOPSIS-based process for IoMT device model update selection. The process begins with normalizing device-provided accuracy values, followed by identifying ideal and worst solutions. Distances to these solutions are calculated, and a closeness coefficient (C_i) is computed for ranking devices. The top-ranked updates are aggregated to form the global model, ensuring contributions from the most reliable devices.

$$S_i^- = \sqrt{\sum_{j=1}^d (w_{ij}^{(t)} - A_j^-)^2}. \quad (12)$$

The relative closeness C_i of each model update to the ideal solution is then calculated as:

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}. \quad (13)$$

The closer C_i is to 1, the more desirable the model update is. The whole process is described in Figure 5.

In the context of similarity assessment, the optimal solution is defined as the highest normalized performance metric (e.g., accuracy) for each client across all devices, indicating the best-performing model, whereas the suboptimal solution is characterized by the lowest normalized performance metric, signifying the worst-performing model. The normalized numbers are utilized to compute the deviation of each device's performance from both the optimal and suboptimal solutions. The closeness coefficient, based on these distances, assesses the degree to which a device's performance corresponds with the ideal standard. This guarantees that the optimization of the global model prioritizes inputs from high-performing devices, hence augmenting the robustness and efficacy of the entire system.

e) *Clustering based on similarity*: The server clusters the model updates into groups based on their similarity C_i values. The clustering process helps in identifying the most similar models, which are then considered for aggregation in the next phase.

E. Outlier Reduction

The calculation of similarity to the ideal solution A^+ and anti-ideal solution A^- directly identifies model updates $\mathcal{M}_i^{(t)}$

Algorithm 1 Federated Learning With TOPSIS and Personalized Updates for IoMT

Require: N IoMT devices, initial global model $\mathcal{M}_G^{(0)}$, number of communication rounds T , learning rate η

Ensure: $\mathcal{M}_G^{(T)}$

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1: Init:  $\mathcal{M}_G^{(0)}$ , secure channels
2: for  $i \in \{1, 2, \dots, N\}$  do
3:    $\mathcal{M}_G^{(0)} \rightarrow D_i$ 
4: end for
5: for  $t \in \{1, 2, \dots, T\}$  do
6:   for  $i \in \{1, 2, \dots, N\}$  in parallel do
7:      $\mathcal{M}_G^{(t-1)} \rightarrow D_i$ 
8:      $\mathcal{M}_i^{(t)} \leftarrow \min \mathcal{L}_i^{\text{reg}}$ 
9:      $\mathbf{w}_i^{(t)} \rightarrow \text{Server}$ 
10:   end for
11:    $\mathbf{w}_i^{(t), \text{norm}} = \frac{\mathbf{w}_i^{(t)}}{\sqrt{\sum_{j=1}^N (\mathbf{w}_j^{(t)})^2}}$ 
12:    $\mathbf{w}_{\text{best}} = \max_i \mathbf{w}_i^{(t), \text{norm}}, \mathbf{w}_{\text{worst}} = \min_i \mathbf{w}_i^{(t), \text{norm}}$ 
13:    $S_i^+ = \sqrt{\sum_{j=1}^d (\mathbf{w}_i^{(t), \text{norm}} - \mathbf{w}_{\text{best}})^2}, S_i^- = \sqrt{\sum_{j=1}^d (\mathbf{w}_i^{(t), \text{norm}} - \mathbf{w}_{\text{worst}})^2}$ 
14:    $C_i = \frac{S_i^-}{S_i^+ + S_i^-}$ 
15:    $\mathcal{M}_G^{(t)} = \sum_{i \in \text{top-rank}} \alpha_i \mathbf{w}_i^{(t)}$ 
16:    $\mathcal{M}_G^{(t)} \rightarrow D_i$ 
17: end for
18: for  $i \in \{1, 2, \dots, N\}$  do
19:    $\mathcal{M}_i^{\text{personal}} \leftarrow \min \mathcal{L}_i^{\text{personal}}$ 
20: end for
21: return  $\mathcal{M}_G^{(T)}, \{\mathcal{M}_i^{\text{personal}}\}_{i=1}^N$ 

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that are consistent with the majority. Therefore outliers, having greater distances and better performances in the client servers S_i^+ to A^+ and smaller distances with performance scores S_i^- to A^- , will result in lower similarity scores C_i . This allows the system to effectively identify and separate outliers from the rest of the data.

F. Robust Aggregation Using Multi-Krum

After dimensionality reduction and clustering, the server employs the Multi-Krum algorithm [57] for robust aggregation to handle adversarial updates while ensuring resilience and reliability in federated learning.

f) Multi-Krum Selection: For each cluster C_k , Multi-Krum evaluates model updates by calculating the pairwise squared Euclidean distances between all client updates. The distance between any two models is computed as the sum of squared differences for each parameter across the model state dictionaries. Using these distances, Multi-Krum assigns a score to each model by summing the distances to its $n - 2f - 2$ closest neighbors, where n is the total number of models, and f is the number of adversarial updates tolerated.

Instead of selecting just one model, Multi-Krum selects the top m models with the smallest scores as representatives

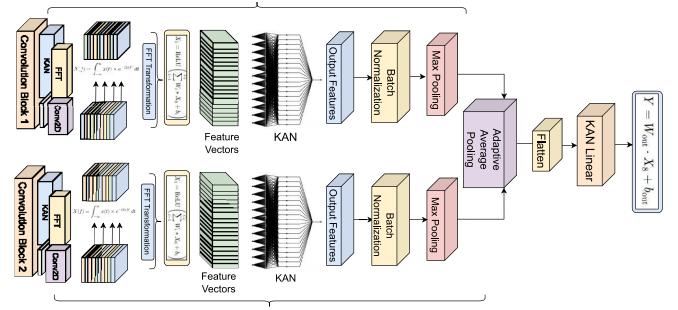


Fig. 6. KANConvNet model architecture for capturing non-linear interactions and processing complex feature transformations in FL.

for the cluster, ensuring that the aggregation incorporates contributions from multiple robust updates.

g) Aggregation: The global model $\mathcal{M}_G^{(t)}$ is updated by averaging the selected models from each cluster:

$$\mathcal{M}_G^{(t)} = \frac{1}{|C|} \sum_{k=1}^{|C|} \frac{1}{m} \sum_{i \in S_k} \mathcal{M}_i^{(t)}, \quad (14)$$

where $|C|$ is the number of clusters, S_k is the set of selected indices for cluster C_k , and $\mathcal{M}_i^{(t)}$ represents the model updates chosen by Multi-Krum for aggregation.

h) Personalized model updates: The final global model is sent back to each device. Personalization is achieved by combining the global model with the device's previous local model:

$$\mathcal{M}_i^{(t+1)} = \alpha \mathcal{M}_G^{(t)} + (1 - \alpha) \mathcal{M}_i^{(t)}, \quad (15)$$

where $\alpha \in [0, 1]$ controls the degree of personalization.

G. KANConvNet Intrusion Detection Model

The KANConvNet model is inspired by the requirement for an efficient and interpretable architecture capable of managing complicated feature interactions in image data, particularly within a federated learning setting. Using the Kolmogorov-Arnold representation theorem [56], the model represents non-linear interactions with tailored activation functions, improving resilience in decentralized training. Its convolutional structure allows scalability and adaptability while processing diverse, large-scale datasets in federated systems. In the model architecture, KAN introduces a novel architecture by relocating activation functions to the edges and applying non-linearity before input summation, fundamentally redefining learning dynamics. This approach enhances interpretability, precision.

The model begins by preprocessing the input vector $\mathbf{x} = [x_1, x_2, \dots, x_{46}]^T$ into $\mathbf{z}_0 = \text{Preprocess}(\mathbf{x})$, using normalization or Fourier transforms. The Kolmogorov-Arnold mapping is then applied, transforming the preprocessed data via a linear transformation and Fourier Transform to $\mathbf{z}_1 = \mathbf{W}_1 \cdot \mathcal{F}(\mathbf{z}_0) + \mathbf{b}_1$. A custom Kolmogorov-inspired activation function is applied, producing $\mathbf{z}_2 = \sigma_{\text{Kolmogorov}}(\mathbf{z}_1)$. The network proceeds through multiple dense layers, refining features with ReLU activations: $\mathbf{z}_3$, $\mathbf{z}_4$, and $\mathbf{z}_5$, each applying distinct weights and biases. The final transformation layer outputs $\mathbf{z}_6 =$

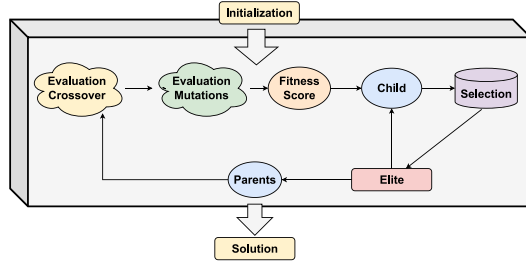


Fig. 7. GA for Hyperparameter Design Space Exploration.

TABLE I
GA TERMINOLOGY

Term	Description
Gene	Represents the value of a dimension in a design point
Genome	A collection of genes that together define a design point
Population	The complete set of all possible genomes in a generation
Elite	A subset of genomes with the highest fitness scores
Mutation	The process of randomly altering selected genes
Crossover	The technique of combining genes from parents to produce offspring genomes

TABLE II
HYPERPARAMETER VALUES

Hyperparameter	Values
Learning rate (α)	{0.00001, 0.0001, 0.01, 0.1, 1.0}
Batch size (β)	{8, 16, 32, 64, 128, 256}
Number of epochs (e)	{1, 5, 20, 50, 100}
Momentum (μ)	{0.1, 0.5, 0.8, 0.99, 0.999}
Dropout rate (δ)	{0.05, 0.1, 0.3, 0.5, 0.7}
Optimizer type (ω)	{adam, sgd, rmsprop, adagrad, nadam}

$\mathbf{W}_5 \cdot \sigma_{\text{Kolmogorov}}(\mathbf{z}_5) + \mathbf{b}_5$, culminating in the prediction $\hat{y} = \text{SGD}(\mathbf{z}_6)$. This architecture effectively captures complex data patterns for accurate predictions.

V. HYPERPARAMETER DESIGN SPACE EXPLORATION

A. Design Space Definition

In our FL-based system, hyperparameter tuning is crucial for optimizing model performance. The design space for our genetic algorithm-based hyperparameter optimization encompasses several key hyperparameters. Let θ represent the set of hyperparameters we aim to optimize. The performance of models is substantially influenced by hyperparameters like as learning rate, batch size, and optimizer type. The learning rate (α) sets the step size for weight updates during training which smaller values provide consistent convergence but slow training, whereas bigger values may produce instability or divergence. Batch size (β) affects gradient estimation accuracy and computational efficiency, where smaller batches introduce noisier gradients that aid in generalization, while bigger batches are computationally faster but may lead to overfitting. The optimizer type (ω) regulates how gradients are applied, effecting convergence speed and stability.

The main and primary hyperparameters included are:

Thus, the design space $\mathcal{H}$ is defined as:

$$\mathcal{H} = \{\alpha, \beta, e, \mu, \delta, \omega\} \quad (16)$$

The values taken across 4 datasets are given in Table II.

B. Search Algorithm

To efficiently navigate the extensive hyperparameter space, we employ a genetic algorithm (GA) [58]. The search algorithm iteratively evolves a population of candidate hyperparameter sets to optimize the performance metric $f(\theta)$. All the keyword descriptions are given in Table I. The process involves several key steps: initialization, evolution through crossover and mutations, and fitness evaluation. The search algorithm is outlined as follows:

C. Genetic Algorithm

1) *Initialization*: The initial population $\mathcal{P}_0$ of size N is generated by sampling hyperparameter sets from the design space $\mathcal{H}$. Each individual $i \in \mathcal{P}_0$ is represented as θ_i :

$$\mathcal{P}_0 = \{\theta_1, \theta_2, \dots, \theta_N\} \quad (17)$$

where

$$\theta_i = \{\alpha_i, \beta_i, e_i, \mu_i, \delta_i, \omega_i, \lambda_i, \gamma_i\} \quad (18)$$

2) *Evolution Crossover*: Crossover combines two parent hyperparameter sets θ_p and θ_q to produce two offspring θ_c^1 and θ_c^2 . The crossover operation is defined as follows:

$$\theta_c^1 = \{\theta_p^k \mid k \in \{1, 2, \dots, 8\}\} \quad (19)$$

$$\theta_c^2 = \{\theta_q^k \mid k \in \{1, 2, \dots, 8\}\} \quad (20)$$

where k indicates the position of each hyperparameter within the set.

3) *Evolution Mutations*: Mutation introduces randomness by altering hyperparameters in the offspring with a small probability P_m . For each hyperparameter θ_i^k in θ_c^1 and θ_c^2 , the mutation operation is:

$$\theta_i^k \leftarrow \begin{cases} \text{random value from } \mathcal{H}_k & \text{with probability } P_m \\ \theta_i^k & \text{with probability } 1 - P_m \end{cases} \quad (21)$$

where $\mathcal{H}_k$ represents the range of possible values for the k -th hyperparameter.

4) *Fitness Score*: The fitness score $f(\theta_i)$ evaluates the performance of the federated learning model trained with hyperparameter θ_i . The score is computed based on the model's accuracy, denoted as A :

$$f(\theta_i) = A(\theta_i) \quad (22)$$

The mutation process promotes variation by periodically adjusting hyperparameter values, helping to explore the search space more fully and avoid local optima. It establishes a balance between exploration and exploitation, enabling the identification of better hyperparameter combinations that could be missed by crossover alone. This finally boosts model performance by fine-tuning important parameters and improving robustness across datasets.

5) *Verification*: The GA process is repeated for several generations G , producing new populations $\mathcal{P}_t$ at each generation t . The population is updated as follows:

$$\mathcal{P}_{t+1} = \{\theta_1, \theta_2, \dots, \theta_N\} \quad (23)$$

Our Algorithm 2 terminates when the improvement in the fitness score falls below a threshold ϵ . We consider the fitness

Algorithm 2 GA Design for Hyperparameter Optimization

```

1: Initialize: Population size  $N$ , number of generations  $G_{\max}$ ,
   mutation probability  $P_m$ 
2: Generate initial population  $\mathcal{P}_0 = \{\theta_1, \theta_2, \dots, \theta_N\}$  where
    $\theta_i \in \mathcal{H}$ 
3: for  $t = 0$  to  $G_{\max} - 1$  do
4:   for each  $\theta_i \in \mathcal{P}_t$  do
5:     Evaluate fitness  $f(\theta_i)$ 
6:   end for
7:   Select top  $N/2$  individuals from  $\mathcal{P}_t$  to form  $\mathcal{P}_t^{\text{sel}}$ 
8:   Crossover:
9:      $\mathcal{P}_{t+1} \leftarrow \emptyset$ 
10:    while  $|\mathcal{P}_{t+1}| < N$  do
11:      Randomly select  $\theta_p, \theta_q \in \mathcal{P}_t^{\text{sel}}$ 
12:      Generate offspring  $\theta_c^1, \theta_c^2$  by:
13:      for each  $k \in \{1, 2, \dots, 8\}$  do
14:         $\theta_c^1[k] \leftarrow \theta_p[k]$  or  $\theta_q[k]$  (randomly chosen)
15:         $\theta_c^2[k] \leftarrow \theta_p[k]$  or  $\theta_q[k]$  (randomly chosen)
16:      end for
17:       $\mathcal{P}_{t+1} \leftarrow \mathcal{P}_{t+1} \cup \{\theta_c^1, \theta_c^2\}$ 
18:    end while
19:    Mutation:
20:    for each  $\theta_i \in \mathcal{P}_{t+1}$  do
21:      for each  $k \in \{1, 2, \dots, 8\}$  do
22:        if random probability  $< P_m$  then
23:           $\theta_i[k] \leftarrow$  random value from  $\mathcal{H}_k$ 
24:        end if
25:      end for
26:    end for
27:  end for
28: Output: Best hyperparameter set  $\theta^* = \arg \max_{\theta_i \in \mathcal{P}_{G_{\max}}} f(\theta_i)$ 

```

TABLE III
BEST HYPERPARAMETERS FOR DIFFERENT DATASETS

Hyperparameter	Wustl-EHMS	EHCL	BlueTrack	RT-LoT
Learning Rate	0.06	0.05	0.03	0.08
Batch Size	16	32	64	16
Epochs	41	50	35	60
Momentum	0.62	0.75	0.52	0.67
Dropout Rate	0.22	0.31	0.19	0.26
Optimizer	SGD	Adam	SGD	SGD
Layer Size	32	64	128	256
Fitness	0.94	0.94	0.95	0.95

score threshold 98. The best hyperparameter set θ^* is selected as:

$$\theta^* = \arg \max_{\theta_i \in \mathcal{P}_G} f(\theta_i) \quad (24)$$

This GA-based hyperparameter optimization in the FedIoMT framework effectively balances exploration and exploitation, to achieve higher model performance with global optima.

VI. EXPERIMENTAL RESULT

A. Experimental Setup

For global testing with our FedIoMT model with our existing works, we compare with FedAvg [51]. FedPer [52],

FedProx [51] model. Besides, for the local testing of our KANConvNet model, we compare it to the SOTA models such as DNN, CNN [53] and SVM [54]. We evaluate our model on the TPU v2, Raspberry Pi 4, TPU VM v3-8, and GPU A100 to assess its performance across diverse hardware configurations.

B. Attack Methods

We assess our model's robustness against various attacks:

- *ALIE*: Injects undetected noise into updates, corrupting them within $(\mu_i - z_{\max} \delta_i, \mu_i + z_{\max} \delta_i)$, where μ_i and δ_i are the mean and standard deviation, respectively, and $z_{\max}$ is from [59].
- *SF Attack*: Flips gradient signs to increase loss, with no need for access to other clients' updates [60].
- *RN Attack*: Adds Gaussian noise to model parameters to degrade performance [61].
- *LF Attack*: Alters class labels in datasets to cause targeted misclassification [62].

1) *Attack Performance Analysis*: In analyzing the robustness of our model with multiple attack types to alter the gradients and labels, we apply those gradient manipulation attacks. The ALIE attack injects noise, SF reverses gradient signs, RN adds Gaussian noise, and LF modifies labels to create misclassification. As demonstrated from Figure 13, the performance with all models along the FedIoMT reduces a little bit. But FedIoMT retains superior performance in validation accuracy in the global model along with another model for handling different adversarial attacks. FedIoMT's principal contribution is in its implementation of the Krum aggregation technique, which improves resilience to adversarial attacks in federated learning. Krum identifies the most dependable gradient updates by assessing the distance between one client's gradient and those of others, excluding outliers. This guarantees that only reliable updates are compiled, notably reducing the effect of gradient manipulation attacks.

C. Impact of Hyperparameters

Hyperparameters in Table III vary among datasets after applying the GA design in our FedIoMT model. The learning rates range from 0.032 to 0.079; batch sizes are 16, 32, or 64; epochs range from 35 to 60. Momentum values range between 0.522 and 0.746. Dropout rates vary from 0.194 to 0.312. Optimizers include SGD, Adam, and RMSprop with fitness scores best for RT-LoT (0.954) and lowest for EHCL (0.936).

D. Superiority of FL Methods for Detecting Attacks

To validate the effectiveness of FedIoMT, we compare it with several existing FL algorithms shown in Table V. The FedIoMT model significantly beats FedPer, FedProx, and FedAvg across all datasets, indicating a notable accuracy increase across all four datasets. For instance, FedIoMT's local final accuracy on Wustl-EHMS is 99.36%, exceeding FedPer by 0.48%, FedProx by 1.13%, and FedAvg by 0.91%. Similar tendencies are observed in global situations, where FedIoMT's accuracy on Wustl-EHMS is 1.38% higher than

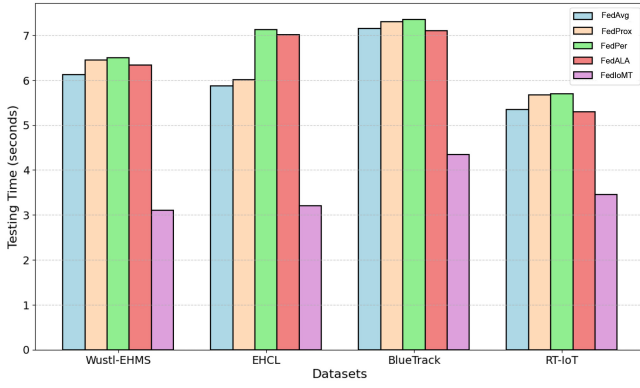


Fig. 8. Measuring Testing Times of four Benchmark IoMT Datasets of our Proposed FedIoMT Model.

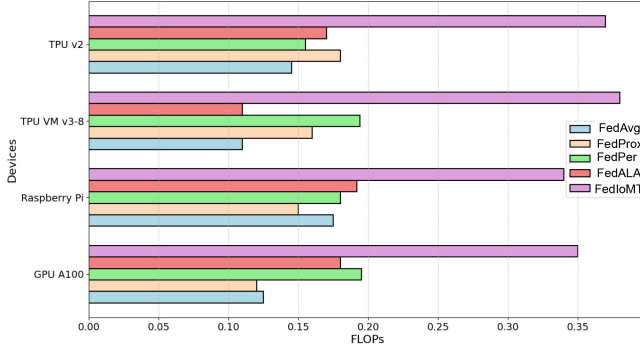


Fig. 9. Measuring FLOP Counts of Different Accelerators along with Edge Devices of our Proposed FedIoMT Model.

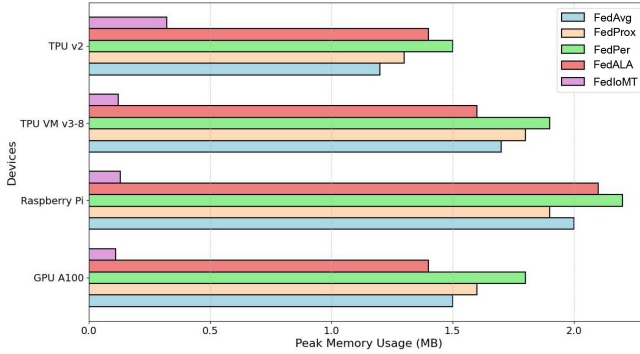


Fig. 10. Measuring Peak Memory Usage of Different Accelerators along with Edge Devices of our Proposed FedIoMT Model.

suggested FedIoMT model, with a very low error rate, as represented in Figure 12.

In Figure 15, t-SNE plots for four distinct datasets, each illustrating various network attack scenarios, to visualize how feature vectors cluster in a two-dimensional space. The FedIoMT model effectively captures low-dimensional data, maintaining critical information in the aggregated updates from local client models.

In terms of assessing the prediction testing time of all the models across the four datasets in Table VIII, FedIoMT consistently displays the shortest testing time, taking 3.10 to 4.35 seconds, which is around 1.5 to 2.3 times faster than other models. In contrast, FedPer records the highest testing

TABLE VIII
COMPARISON OF KRUM AND MULTI-KRUM RESULTS

Metric	Krum	Multi-Krum
Average Aggregation Time	0.0016 seconds	0.0008 seconds
Final Accuracy	0.9997	0.9997
Average Evaluation Time	0.0033 seconds	0.0023 seconds

time, especially on the Wustl-EHMS dataset, where it is nearly 1.7 times slower than FedIoMT. This illustrates FedIoMT's efficiency in decreasing testing time across diverse datasets.

E. Ablation Study

The ablation studies performed on our proposed FedIoMT architecture indicate its robust performance and efficiency across major components.

The first study shown in Table VII assesses the impact of adding the TOPSIS algorithm for client-level feature ranking. The results reveal consistent accuracy improvements across most clients, with gains ranging from +0.0003 to +0.0004. This emphasizes the effectiveness of TOPSIS in increasing model performance by prioritizing essential features during training.

The second study displayed in Table VIII examines the Krum and Multi-Krum aggregation methods within the FedIoMT framework. While both approaches reach the same high-end accuracy of 0.9997, Multi-Krum far surpasses Krum in computing efficiency. Multi-Krum lowers the aggregation time by 50% and the evaluation time by approximately 30%, making it a scalable and low-latency alternative for real-time or resource-constrained applications. Therefore, these investigations provide the efficacy of the FedIoMT architecture in boosting both accuracy and efficiency, revealing its promise for secure and scalable federated learning.

F. Advantage of KANConvNet

We compare the performance of FedIoMT with different DL models to investigate the effectiveness of KANConvNet in Table IV. The performance measurements indicate that CNN regularly surpasses the other models in local settings, attaining the highest level of accuracy, reaching a peak of 99.58% on Wustl-EHMS. KANConvNet closely follows, exhibiting a marginal drop of 0.14% to 0.46% compared to CNN, but generally surpassing the performance of DNN and SVM models. The DNN model has a strong performance on the Wustl-EHMS dataset, achieving an accuracy of 99.28%. However, it falls slightly below the CNN model by approximately 0.30%. The Support Vector Machine (SVM) has the lowest local accuracy, reaching a maximum of 98.89%, which is a drop of around 0.69% compared to the CNN. During global testing scenarios, all models exhibit a decrease in accuracy. However, KANConvNet has a higher level of performance (97.90% on Wustl-EHMS) compared to DNN and SVM, which decrease to 96.67% and 96.51% respectively. These variances among datasets emphasize the durability of KANConvNet's performance in both local and global scenarios, while DNN and SVM lag in global generalization, notably in the RT-IoT dataset. Furthermore, in 3D t-SNE visualization,

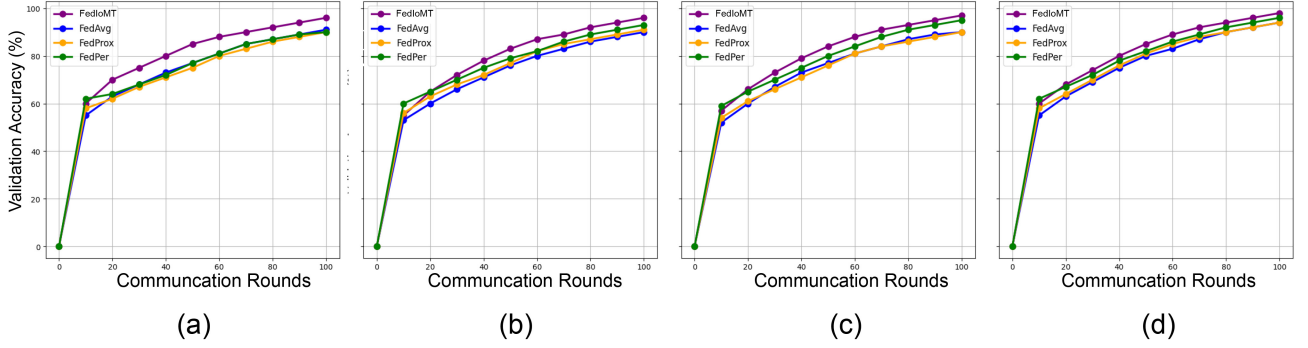


Fig. 11. Global Validation Accuracy of different algorithms along with FedIoMT model for (a) Wustl-EHMS (b) EHCL (c) BlueTrack (d) RT-IoT Datasets.

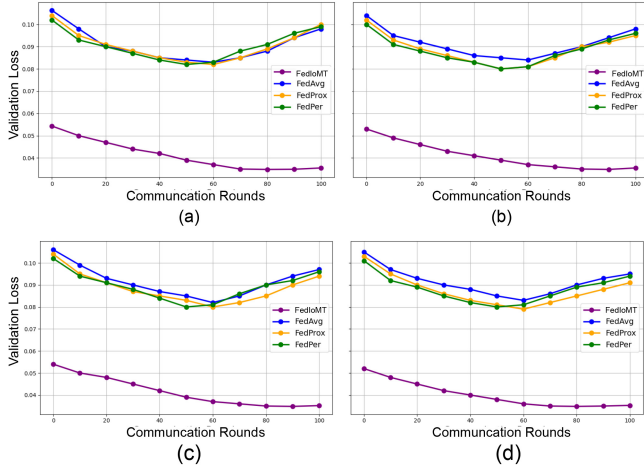


Fig. 12. Global Validation Loss of different algorithms along with FedIoMT model for (a) Wustl-EHMS (b) EHCL (c) BlueTrack (d) RT-IoT Datasets.

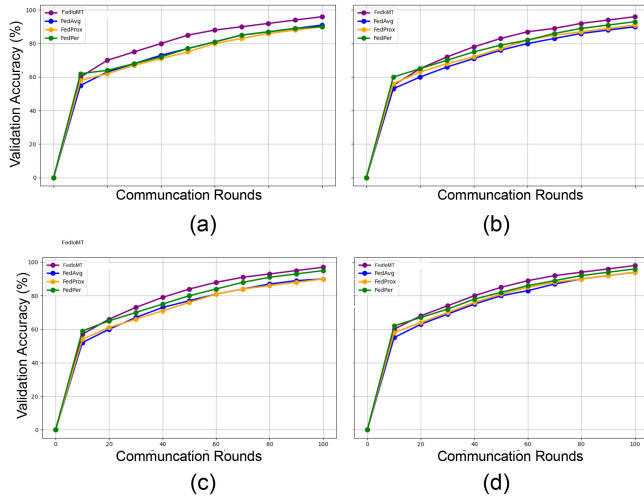


Fig. 13. Global Validation Accuracy of different algorithms along with FedIoMT model for (a) Wustl-EHMS (b) EHCL (c) BlueTrack (d) RT-IoT Datasets after applying four attack types.

the KANConvNet model is shown independently and looks to have a smoother accuracy curve than other models, indicating that our KANConvNet model performs rather consistently across communication rounds shown in Figure 14.

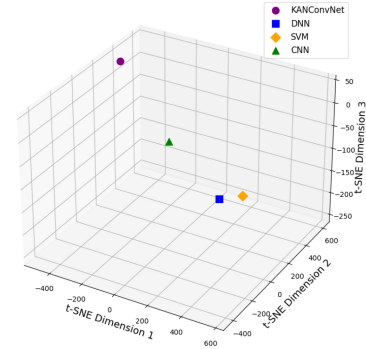


Fig. 14. 3D t-SNE visualization of KANConvNet model as local classifier along with other SOTA models.

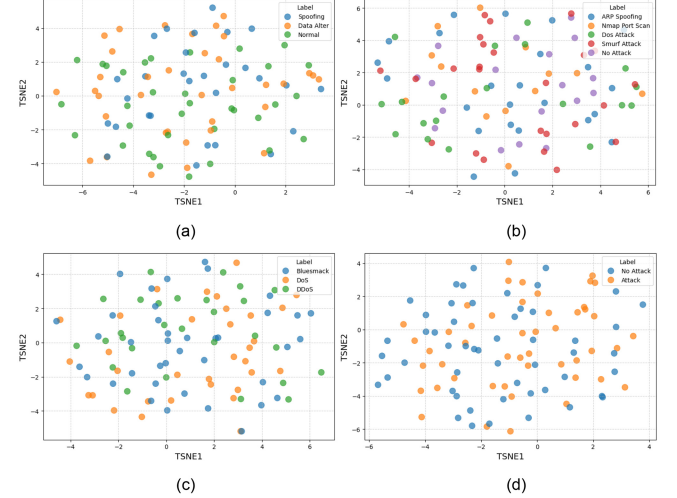


Fig. 15. t-SNE plots of four datasets illustrating network attack scenarios, showing how the FedIoMT model preserves critical low-dimensional data in aggregated client updates (a) Wustl-EHMS (b) EHCL (c) BlueTrack (d) RT-IoT datasets.

G. Training Efficiency on Hardware Accelerators

The FLOP mainly counts comparison across different devices and models reveals notable variations in computational efficiency. Overall, FedIoMT consistently shows superior computational efficiency across all devices, with value counts of 0.24-0.29 in Table IX. In those diverse configuration devices, the Raspberry Pi shows the largest peak memory utilization, up to 2.2 MB with FedPer, almost 1.8 times greater than the

TABLE IX
PERFORMANCE COMPARISON WITH AVAILABLE WORKS

Model	Accuracy	Recall	Meta Learning
FedCNNMLP [66]	95.85%	93.92%	✗
FL-MGVN [67]	92.65%	92.78%	✗
Mamunur et al. [45]	91.12%	97.89%	✗
Priyanka et al. [46]	99.32%	99.65%	✗
Farhan et al. [68]	99.24%	-	✗
FedIoMT (Ours)	99.43%	99.53%	✓

GPU A100's 1.5 MB. The TPU v2 has the lowest memory use, nearly 2.9 times lower than the Raspberry Pi's peak. FedIoMT continuously utilizes the least RAM across all devices showcasing its efficiency shown in Table IX.

H. Discussion

The performance of our proposed FedIoMT model is compared with other state-of-the-art methods in Table IX, highlighting its superiority in both accuracy and recall. FedIoMT has an accuracy of 99.43% and a recall of 99.53%, outperforming the FedCNNMLP, FL-MGVN, and Mamunur et al. models on both criteria. At the same time, Priyanka et al. et al. and Farhan et al. models depict superior performance in recall metrics than our FedIoMT model. However, these existing studies have the disadvantage of securing efficient communication overhead throughout the detection time of attack types. Furthermore, these works lack the issue of maintaining an accurate weights update mechanism whereas our proposed FedIoMT architecture overcomes the issues of decentralized multi-center servers by using TOPSIS similarity analysis to reduce data heterogeneity during local-to-global transmission. Furthermore, in the local model, our KANConvNet uses Kolmogorov activation functions at the edges rather than the neuron's core, which may alter the learning dynamics and improve the overall interpretability of the FedIoMT framework.

VII. CONCLUSION

The IDS played a crucial role in securing IoMT devices and networks within consumer electronics, addressing unique challenges and high risks. However, many existing methodologies that relied on machine learning (ML) and deep learning (DL) techniques were considered opaque by security analysts and engineers. The proposed FedIoMT framework addressed data heterogeneity and model adaptation through multi-center clustering and meta-learning. The innovative KANConvNet model, based on the KAN architecture, enhanced scalability, interpretability, and intrusion detection by capturing spatial and hierarchical data. The FedIoMT system also improved security through a meta-learning phase and TOPSIS-based model aggregation, while optimizing hyperparameters using Genetic Algorithms. Despite its advantages, FedIoMT faced challenges in managing severe heterogeneity in real-world IoMT datasets, particularly when devices exhibited significantly outlier data distributions. This issue reduced performance, even with multi-center clusters and similarity analysis. Additionally, the meta-learning step, while augmenting model robustness, imposed computational burdens due to the need to evaluate inference times during training. Although

the Kolmogorov-Arnold theorem proficiently supported specific functions like intrusion detection, its applicability to various IoMT applications, including predictive maintenance and real-time monitoring, required further evaluation. Future work aimed to optimize computational efficiency for low-power consumer devices, enhance scalability, improve privacy protections, and refine dimensionality reduction techniques. Efforts to prevent overfitting and ensure seamless deployment across diverse IoMT devices were also essential.

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